

MEMORANDUM

SUBJECT: Standard Assumptions for PMN Assessments

FROM: Cathy, Gail, Nhan, Scott, Kin

TO: All CEB Members

Attached is a chart summarizing the assumptions for releases of liquid residue from unloading and clean-up operations for batch operations. Also, listed below are some other assumptions and/or generic language which all of us should be aware of.

Baghouse/Cyclone: use 99% efficiency

Carbon absorption: use 50% efficiency

Unreacted Intermediate: if submitter claims the PMN is “completely reacted”, assume that it is 99% reacted. Also, for IRER, we need to provide a qualitative statement about the potential exposures and releases of the PMN downstream. An example of this qualitative language is:

“Should this PMN not be fully reacted in use, residuals in reaction masses may result in exposures and release in later processing/use. {One example would be that commercial painters may receive potential doses from inhalation of vapor and mists during handling and application of paints containing residuals.}”

Release from drum residue for solids: in the interim pending future experimental investigation (e.g. for liquids), use:

- 1% by weight of drum for dry material and
- 4% by weight of drum for wet material (e.g. wet cake)

Please start using these assumptions for estimating releases of PMNs so that our IRERs will be more consistent.

Suggested Assumptions for Release of Liquid Residue from Unloading and Clean-Up Operations for Batch Operations

Releases from clean-up = f (clean-up frequency, unloading method, # of unit operations in process, solubility of material in water/organic solvent, cleaning process, viscosity, type and amount of solvent, and lining of containers).

* We will not take into account the following factors in estimating release of residue from unloading and clean-up operations: lining of container, viscosity, type and amount of solvent, and cleaning method.

Key: RQ = Release Quantity RF = Release Frequency CM = Cleaning Method RD = Release Disposition

Issue: Single vessel versus entire process

- The single vessel release is applicable of processes involving one-unit operation such as reaction. Examples of this type of processes include: adhesives formulation, one-step polymerization processes, repackaging (into smaller containers) of an imported product. The engineer should use the release from an entire process if in doubt.





